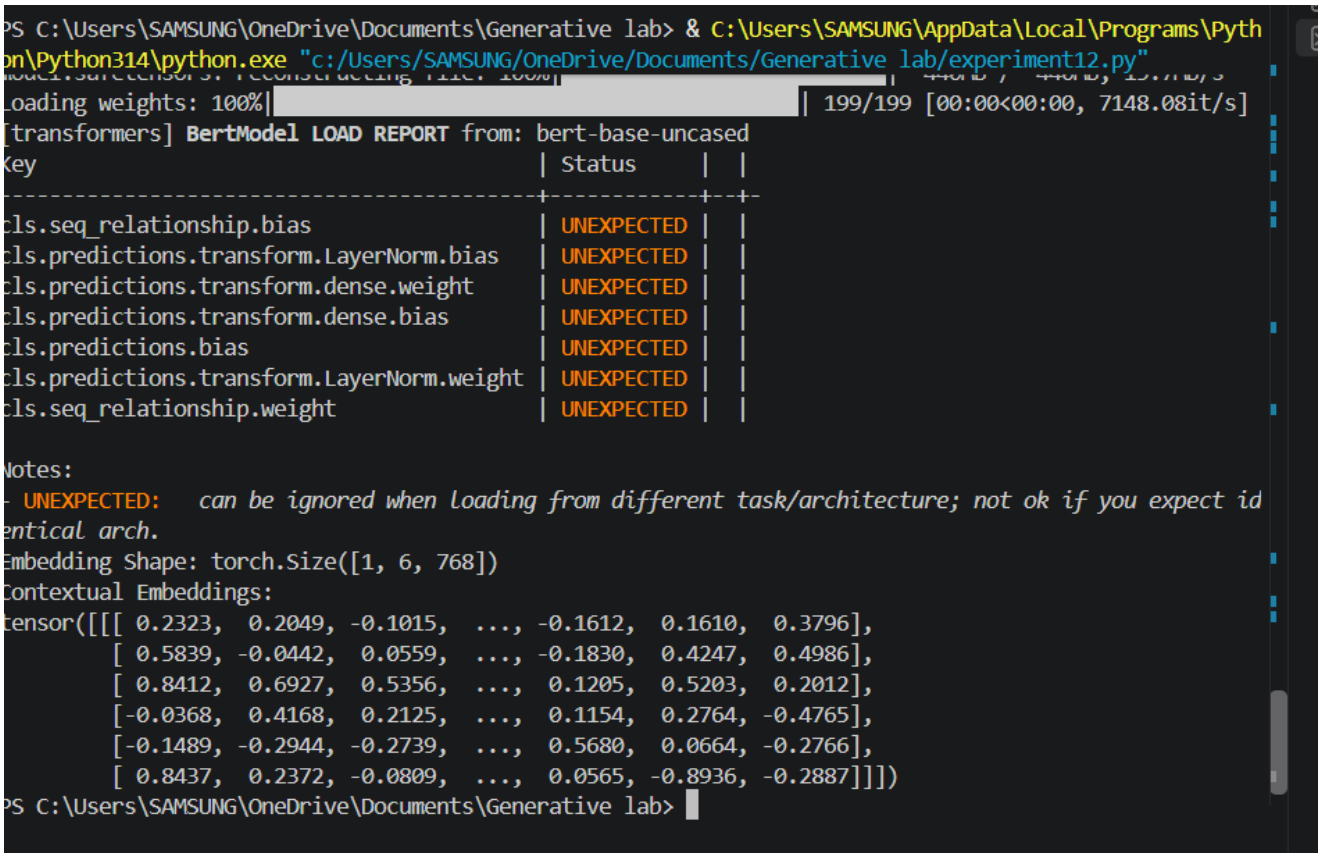
Experiment 12: Load a Pre-trained BERT Model and Generate Contextual Embeddings

Python Program

```
import torch
from transformers import BertTokenizer, BertModel

tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
model = BertModel.from_pretrained("bert-base-uncased")

text = "I love learning Python"
inputs = tokenizer(text, return_tensors="pt")

with torch.no_grad():
    outputs = model(**inputs)

print("Embedding Shape:", outputs.last_hidden_state.shape)
print("Contextual Embeddings:")
print(outputs.last_hidden_state)
```

Output


PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab> & C:\Users\SAMSUNG\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/SAMSUNG/OneDrive/Documents/Generative lab/experiment12.py"
Loading weights: 100%| | 199/199 [00:00<00:00, 7148.08it/s]
[transformers] BertModel LOAD REPORT from: bert-base-uncased
Key | Status | |
cls.seq_relationship.bias | UNEXPECTED | |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED | |
cls.predictions.transform.dense.weight | UNEXPECTED | |
cls.predictions.transform.dense.bias | UNEXPECTED | |
cls.predictions.bias | UNEXPECTED | |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED | |
cls.seq_relationship.weight | UNEXPECTED | |
Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Embedding Shape: torch.Size([1, 6, 768])
Contextual Embeddings:
tensor([[[0.2323, 0.2049, -0.1015, ..., -0.1612, 0.1610, 0.3796],
[0.5839, -0.0442, 0.0559, ..., -0.1830, 0.4247, 0.4986],
[0.8412, 0.6927, 0.5356, ..., 0.1205, 0.5203, 0.2012],
[-0.0368, 0.4168, 0.2125, ..., 0.1154, 0.2764, -0.4765],
[-0.1489, -0.2944, -0.2739, ..., 0.5680, 0.0664, -0.2766],
[0.8437, 0.2372, -0.0809, ..., 0.0565, -0.8936, -0.2887]]])
PS C:\Users\SAMSUNG\OneDrive\Documents\Generative lab>